

The user and system requirements provide a detailed foundation for the flower shop system . However,

the project's overall score is heavily penalized due to major errors in the scheduling calculations, which led to unrealistic timeline predictions Recalculat the schedule based on a parallel team framework and detailing the meeting minutes will significantly



Faculty of Engineering & Technology

Department of Electrical & Computer Engineering.

COMP433, SOFTWARE ENGINEERING

Phase Two:

Requirement Engineering- Elicitation & Discovery

Prepared By:

Marwa Faqeeh 1220039

Roa Makhtoob 1221636

Shaymaa Dar Taha 1222440

Abeer Hussein 1220425

Instructor: Dr. Mohammed Husseinali

Section: 1.

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Task 2.1: USER REQUIREMENTS

Table 1: Task 2.1 Leadership and Contributions

Role	Name	Contribution
Lead	Roa Makhtoob	Writing and coordination
Contributor	Abeer Hussein	Reviewing
Contributor	Shaymaa Dar Taha	Discussion & coordination
Contributor	Marwa Fageeh	Feasibility feedback & review

UR1: Fresh flower browsing

The system shall allow customers to browse fresh flowers, including individual bouquets, seasonal selections, and imported flowers.

UR2: Order placement

The system shall allow customers to place orders for fresh flowers and pre-designed floral arrangement packages.

UR3: Custom arrangement design

The system shall allow customers to customize floral arrangements based on occasion, preferences, and budget, with the option to request on-site setup or installation.

Physical venue installation represents a separate operational field service that should be isolated from digital ordering logic

UR4: Delivery services

The system shall provide local delivery services within the city, including a same-day delivery option subject to an additional fee .

UR5: Subscription services

The system shall allow customers to subscribe to a recurring flower delivery service with flexible weekly, monthly, or yearly plans and customizable schedules.

UR6: Scheduled delivery

The system shall allow customers to schedule delivery for a future date and time.

UR7: Payment methods

The system shall support multiple payment methods, including cash (in-store), credit cards, and online payments.

UR8: Consultation booking

The system shall allow customers to book consultation sessions (in-store or virtual) for floral design assistance.

UR9: Workshop registration

The system shall allow customers to register for floral workshops (online or in-store) at beginner, intermediate, or advanced levels.

UR10: Customer account management

To prevent integration problems, these features should be broken down into separate user requirements

The system shall allow customers to create and manage a personal account to view order history, track deliveries, manage subscriptions, and save personal preferences, delivery addresses and submit reviews and ratings.

different services, separate them

Grade: B

Task 2.2: SYSTEM REQUIREMENTS

Several req specify architectural details, data replication schemes, or client platform limits that belong in an environmental or non-functional requirement specification

Table2: Task 2.2 Leadership and Contributions

Role	Name	Contribution
Lead	Abeer Hussein	Writing and coordination
Contributor	Roa Makhtoob	Requirement alignment
Contributor	Marwa Fageeh	Technical validation
Contributor	Shaymaa Dar Taha	Coordination & finalising

UR1: Fresh flower browsing

SR1.1: The system shall display flowers organized into categories such as bouquets, seasonal flowers, and imported flowers.

SR1.2: The system shall display flower details including name, price, image, description, and availability, and allow customers to view full details of a selected flower.

SR1.3: The system shall allow customers to search for flowers by name or keywords.

SR1.4: The system shall allow customers to filter flowers based on category, price range, and occasion.

SR1.5: The system shall indicate whether a flower is available or out of stock.

SR1.6: The system shall allow staff to assist customers in browsing and viewing flower details in-store

UR2: Order placement

SR2.1: The system shall allow customers to add flowers or arrangements to a shopping cart (online orders).

SR2.2: The system shall allow staff to create orders for customers in-store.

SR2.3: The system shall allow customers or staff to update or remove items from the order.

SR2.4: The system shall display the total price of the order.

SR2.5: The system shall allow customers or staff to confirm and submit an order.

SR2.6: The system shall generate a unique order ID for each confirmed order.

SR2.7: The system shall store order details including items, quantities, and total cost to prevent data loss during system failure.

Specifying database recovery rules, introduces an architectural fault-tolerance mechanism rather than a core functional requirement

SR2.8: The system shall ensure that an order contains at least one item before confirmation.

SR2.9: The system shall confirm and process orders within 3 seconds under normal operating conditions.

UR3: Custom arrangement design

SR3.1: The system shall allow customers to enter and select details for a custom arrangement, including occasion, preferences, colors, budget, and desired flower types or styles.

SR3.2: The system shall allow customers to request on-site setup or installation.

SR3.3: The system shall store custom arrangement requests for processing by the florist

SR3.4: The system shall allow staff to review and manage custom arrangement requests.

SR3.5: The system shall allow staff to update the status of custom orders (pending, confirmed, completed).

SR3.6: The system shall allow staff to provide confirmation or feedback for custom arrangement requests.

UR4: Delivery services

SR4.1: The system shall allow customers or staff to enter delivery details including address, contact information, and delivery instructions.

SR4.2: The system shall validate that the delivery address is within the supported delivery area.

SR4.3: The system shall allow customers or staff to select delivery type (standard or same day).

SR4.4: The system shall apply an additional fee for same-day delivery.

SR4.5: The system shall store delivery details with the order.

SR4.6: The system shall allow staff to manage and update delivery status (scheduled, out for delivery, delivered).

UR5: Subscription services

SR5.1: The system shall allow customers to select a subscription delivery frequency (weekly, monthly, or yearly).

SR5.2: The system shall allow customers or staff to define the subscription duration, including start date and end date.

SR5.3: The system shall allow customers or staff to customize subscription delivery schedules, including preferred delivery dates, times, addresses, and flower preferences.

SR5.4: The system shall allow customers to choose a subscription tier (Basic or Premium).

SR5.5: The system shall assign and automatically apply subscription benefits based on the selected tier, including discounts, delivery advantages, priority processing, and access to seasonal or exclusive flowers.

SR5.6: The system shall allow customers or staff to pause, modify, renew, or cancel a subscription.

SR5.7: The system shall store subscription details including delivery frequency, duration, schedule, preferences, and tier.

SR5.8: The system shall allow staff to view and manage customer subscriptions.

SR5.9: The system shall notify customers about upcoming subscription deliveries and subscription expiration dates

UR6: Scheduled delivery

SR6.1: The system shall allow customers or staff to select a future delivery date.

SR6.2: The system shall validate that the selected delivery date is available for delivery.

SR6.3: The system shall store scheduled delivery details with the order.

SR6.4: The system shall allow staff to view and manage scheduled deliveries.

UR7: Payment methods

SR7.1: The system shall allow customers or staff to select a payment method for an order.

SR7.2: The system shall support cash payment for in-store orders.

SR7.3: The system shall support credit card payments.

SR7.4: The system shall support online payment methods.

SR7.5: The system shall confirm payment before completing an order.

SR7.6: The system shall store payment details associated with each order. limits technical design choices early in the process

SR7.7: The system shall store payment information securely in encrypted form associated with each order.

SR7.8: The system shall ensure secure payment transactions using encrypted

UR8: Consultation booking

SR8.1: The system shall display available consultation time slots.

SR8.2: The system shall allow customers or staff to select a consultation type (in-store or virtual).

SR8.3: The system shall allow customers or staff to select a consultation date and time.

SR8.4: The system shall validate that the selected time slot is available.

SR8.5: The system shall store consultation booking details including date, time, and type.

SR8.6: The system shall allow staff to view and manage consultation bookings.

UR9: Workshop registration

SR9.1: The system shall display available workshops including level, date, time, and price.

SR9.2: The system shall allow customers or staff to select a workshop type (online or in-store).

SR9.3: The system shall allow customers or staff to register for a selected workshop.

SR9.4: The system shall validate that workshop capacity is not exceeded before registration.

SR9.5: The system shall store workshop registration details including participant information and selected workshop.

SR9.6: The system shall allow staff to view and manage workshop registrations.

UR10: Customer account management

SR10.1: The system shall allow customers to create and log in to an account.

SR10.2: The system shall allow customers to view their order history.

SR10.3: The system shall allow customers to track the status of their deliveries.

SR10.4: The system shall allow customers to manage their subscription settings.

SR10.5: The system shall allow customers to save and update delivery addresses and personal preferences.

SR10.6: The system shall allow customers to submit ratings and reviews for completed orders.

SR10.7: The system shall store customer account information securely.

SR10.8: The system shall allow staff to view customer account details when needed.

SR10.9: The system shall support access through modern web browsers including Chrome, Firefox, Edge, and Safari. those are environmental constraints, not functional account features

SR10.10: The system shall maintain customer account data availability with regular backup procedures.

Task 2.3: EFFORT&TIME ESTIMATION

Table2: Task 2.3 Leadership and Contributions

Role	Name	Contribution
Lead	Shaymaa Dar Taha	Writing and coordination
Contributor	Abeer Hussein	System complexity estimation
Contributor	Marwa Faqeeh	Development effort estimation
Contributor	Roa Makhtoob	Scope clarification & reviewing

Acronyms: **pw** = person week, **pm** = person month, **w** = week, **m** = month.

Assumptions:

1 week = 5 days

1 working day = 8 hours

Average developer salary =\$250 per week

Schedule time includes an additional 30% buffer for testing, meetings, fixes, and unexpected

you calculated a minimum duration of 35 calendar weeks and a maximum of 88 calendar weeks to build a simple e-commerce flower website . This timeline means the project would take up to 2 years to finish

UR	Estimated No of Developers	Estimated Effort (Average/developers)	Total Effort (for one Developer)
UR1 – Fresh flower browsing	2	2 pw	$2*2= 4$ pw
UR2 – Order placement	3	3 pw	$3*3= 9$ pw
UR3 – Custom arrangement design	2	3 pw	$2*3= 6$ pw
UR4 – Delivery services	2	2 pw	$2*2= 4$ pw
UR5 – Subscription services	3	4 pw	$3*4= 12$ pw
UR6 – Scheduled delivery	2	2 pw	$2*2= 4$ pw
UR7 – Payment methods	3	3 pw	$3*3= 9$ pw
UR8 – Consultation booking	2	2 pw	$2*2= 4$ pw
UR9 – Workshop registration	2	2 pw	$2*2= 4$ pw
UR10 – Customer account management	3	4 pw	$3*4= 12$ pw
Total: 10	24	27 pw	68 pw

the scheduling math completely ignores your team size. A 4-developer pool works concurrently on parallel tasks . Instead of dividing the total effort (68 person-weeks) by the 4 available developers to find the active calendar period, the team treated person-weeks as a raw calendar baseline

Schedule Time Estimation:

Minimum Completion Time Using average effort: $27 * 1.30 = 35$ weeks

Maximum Completion Time Using total effort: $68 * 1.30 = 88$ weeks

Cost Estimation:

Estimated project cost: $250 * 88 = \$22000$

Profit Margin Estimation:

Minimum Cost with 10% Profit = $22000 * 1.10 = \$24200$

Maximum Cost with 30% Profit = $22000 * 1.30 = \$28600$

The project cost (\$22,000) was calculated by multiplying the developer salary (\$250) by the *calendar time metric* (88 weeks) instead of multiplying by actual effort units. This means you are paying a single developer for 88 weeks, which does not match your 4-person team structure

Table3: Meeting Management Leadership and Contributions

Role	Name	Contribution
Lead	Marwa Faqeeh	Document Writing & Customer Meeting Management

Grade : C+

Minutes of meetings:

An online meeting was held outside class hours to discuss the user and system requirements for Phase 2. The developer and customer groups reviewed the project requirements, exchanged feedback, and finalized the main system features.

- **Date:** 2/5/2026
- **Time:** 9:00 PM – 9:45 PM
- **Attendees:**

Only One Written Session Logged

Developer Group:

Marwa Faqeeh	1220039
Roa Makhtoob	1221636
Shaymaa Dar Taha	1222440
Abeer Hussein	1220425

Customer Group:

Abdallah Kokash	1220116
Bilal Ashhab	1220307
Hanna Kaibni	1220214
Jamil Habash	1220194

- **Topic of Discussion:**

- Overall flower shop workflow from flower selection to order delivery.
- Delivery services, including same-day and scheduled delivery options.
- Payment methods and online payment support.
- Consultation booking services (in-store and virtual).
- Workshop registration and subscription services.
- Customer account management and delivery tracking features.
- Review and clarification of user and system requirements for Phase 2.

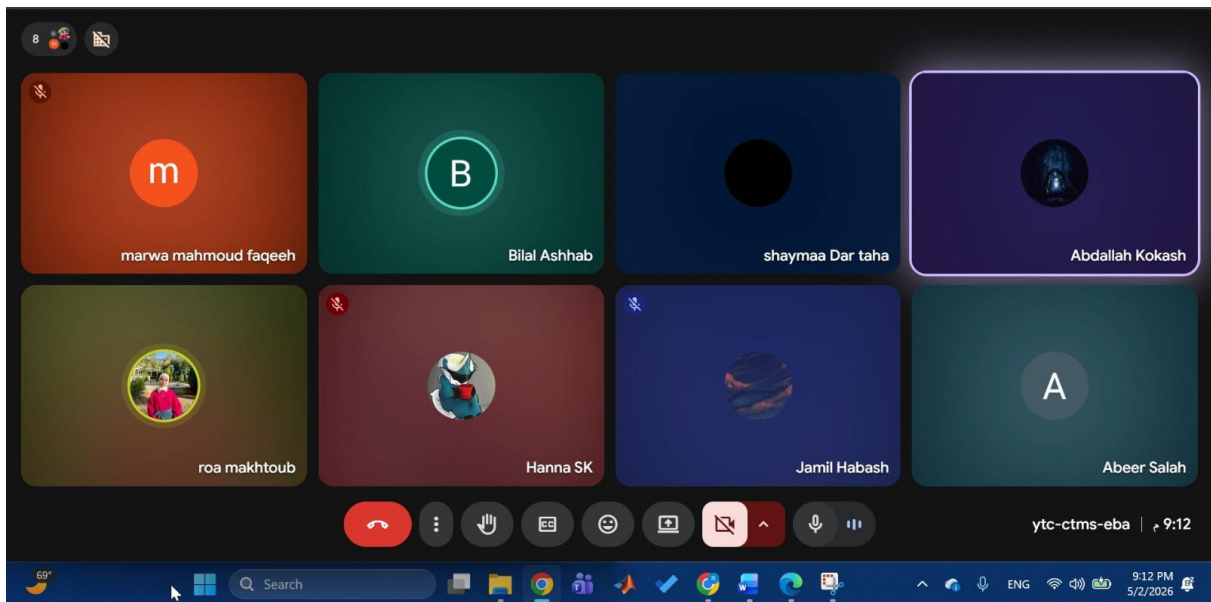


Figure 1: Online Meeting with Customer Group

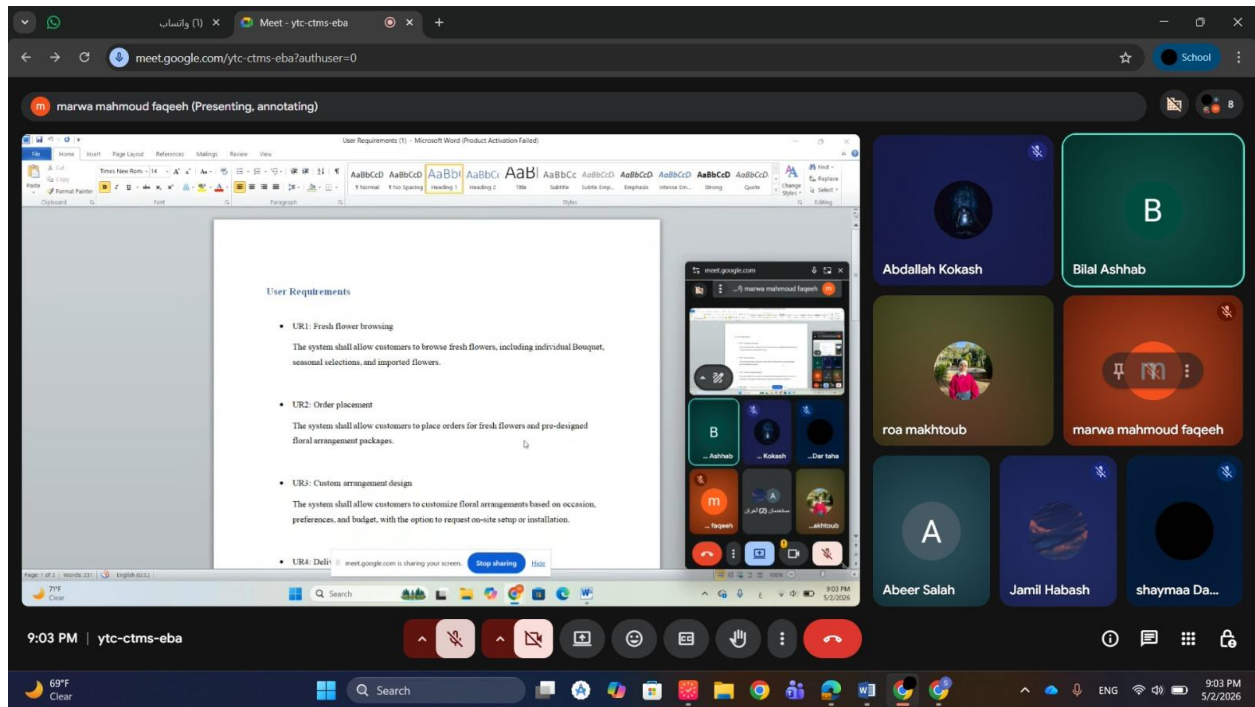


Figure 2: Requirements Discussion Meeting

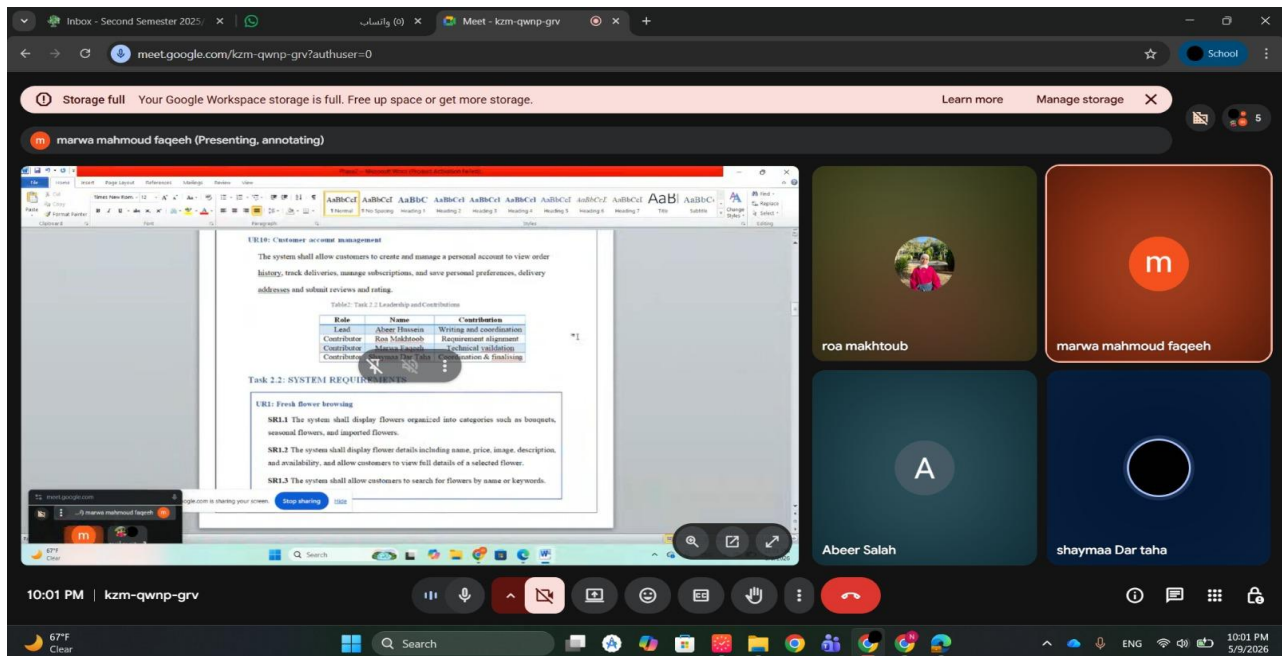


Figure 3: Team Meeting to Finalize the Task

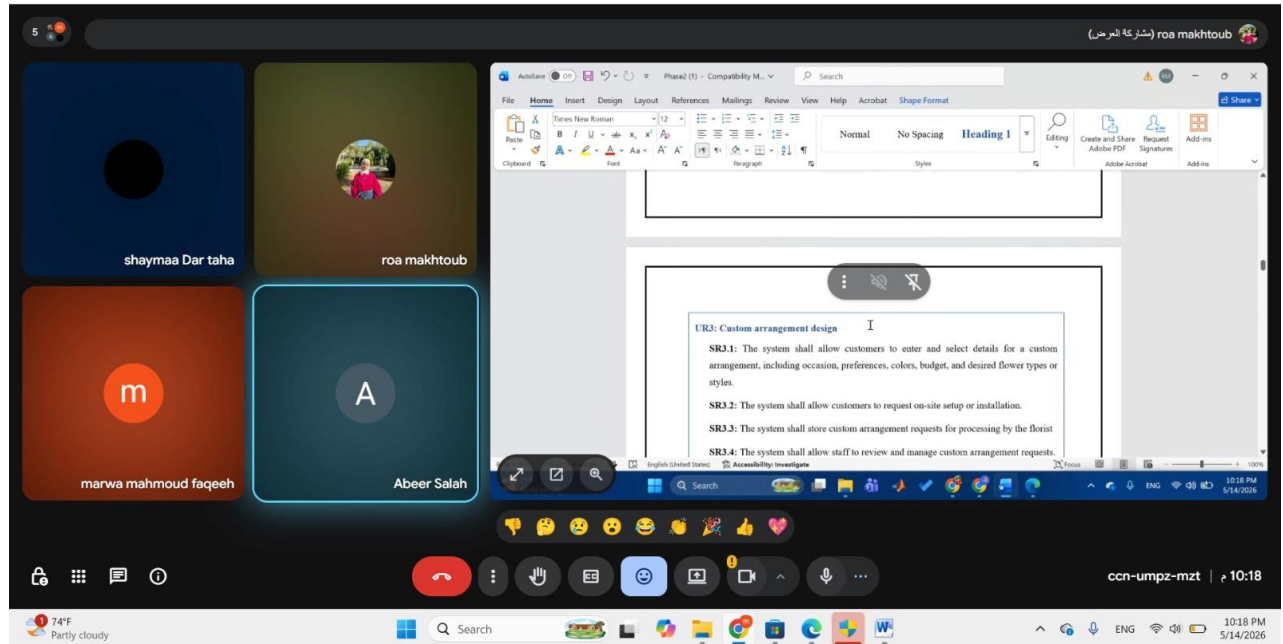


Figure 4: Discussion of the Feedback on Phase 1.1